

WMS Build Roadmap

Warehouse Management System — build pipeline snapshot — as of 2026-09-15

Master Data	Yard & Gate	Inbound	Putaway	Inventory	Outbound	Picking	Dispatch	Returns	Analytics
Built	Built	Built	Built	Built	Started	Started	Not started	Not started	Started

What's landed since the last refresh (2026-09-15)

A Picking gap-fix pass — 4 of the 6 items flagged in the 2026-09-14 gap-analysis table (page 5) now have real, verified code behind them, on top of that session's already-migrated schema.

Short-stock retry + Pick Face wiring — built & verified

A shared `attemptSourceRecovery()` path (never deletes a task/trip — trims quantity, spins off a sibling `NEEDS_SOURCE` task for whatever's unmet) backs a new "Retry Source" action, and now also auto-logs a `SHORT_STOCK` exception the moment a task first goes short, not just on retry. Separately, `suggestPickPosition()` is now a two-phase wrapper — Pick Face (SPR) is always tried before reserve storage, matching Putaway's own pattern. Pick Face's fast-access A/B stock is finally consulted by real picking.

Operator-assignment fairness for Picking — built & verified

Mirrors Putaway's own fairness layer almost verbatim: idle-time ranking among MHE-capable operators, a live recommendation banner, and a new scheduler that alerts a Supervisor — then escalates to a Manager — if the same missed turn isn't picked up. Its own Company Settings grace-minutes dial, separate from Putaway's.

Mid-pick exception handling — designed & built same session

Was flagged "needs its own design pass, not yet started" as of 2026-09-14. Short quantities are now auto-detected on completion; an operator can also explicitly report Empty Bin / Wrong SKU / Damaged / Other via a new "Report a Problem" action, which abandons the trip and recovers a new source. Supervisor+ get a review queue.

Verification

A throwaway-company diagnostic script invoking the real, unmodified service methods directly — 23/23 checks passing. Backend + frontend both compile clean. A live browser click-through was skipped this pass (another session held the dev-server ports) — worth doing before fully trusting the UI side.

Still open — next up

Claim-expiry for an abandoned `PickTrip` (Putaway already solved this exact problem) and a **Picking metric in Analytics** remain deferred, per plan — then **Dispatch**, the real next module in the build order.

Module-by-module status

Module	Status	Summary
Master Data	Built	Warehouses (network-node master: Factory/DC/CNF/Cross-dock), SKUs, Customers, Locations (with a full 3D Plan View), Users — all with role-scoped access and standard import/export.
Yard & Gate	Built	Gate In/Out, dock assignment, document checks, real compass-direction dock zones + a live-rotating 3D compass overlay. One competitor-research pass done (self-check-in + physical inspection flagged, not built).
Inbound	Built	Order maker + Excel bulk import + ERP push, vehicle-matched orders, scan-based receiving, Complete Inward Process/Dock Out, per-dock default staging locations.
Putaway	Built	Real bin-suggestion logic for every storage type (Rack/SPR, Drive-in, Ground/Floor, Stillage) — ABC/FMS-aware, reservation-aware, dock-relative, and (Rack + Stillage) row/rank-aware. Pallet consolidation, Pick Face for SPR, operator-assignment fairness, Aisle/Level Rank visuals. Open: cancel path, queue ordering, Pick Face for Drive-in/Ground/Stillage, Stillage's "closed column" lifecycle.
Inventory	Built	On-hand Line Items + SKU Summary views, a transaction-level ledger, and Excel export on all three views (with From/To Location) — all live-derived, no stored counters. The Daily Volume dashboard's outward half is a separate, still-open Analytics item.
Outbound	Started	Order maker built (manual creation, candidate-vehicle matching by destination city, Supervisor+ allotment reserving real pick tasks at real on-hand). Not built: Excel import, ERP push, vehicle weight/volume capacity check.
Picking	Started	Core execution: position-scoped dynamic source resolution (lane-aware, front-most, aging-bucket FIFO), FIFO trip claiming, two-tier scan completion, real PICK/PICK_IN ledger. 2026-09-15: short-stock retry, Pick Face wiring, operator-assignment fairness, and mid-pick exception handling (auto short-quantity + explicit report) all now built & verified — see page 5. Still open: claim-expiry for an abandoned trip, a Picking metric in Analytics. Wave/batch/ zone/cluster modes and Task Interleaving with Putaway remain deliberately deferred.
Dispatch	Not started	Schema exists (isPriority gate, Dispatch model). No design conversation has happened yet — the next real piece of this trio.
Returns	Not started	Nothing built yet.
Analytics	Started	Operator productivity (marrying/putaway time per operator, Pick Face trip time, abandoned-claim flags) plus Daily Inward Volume charts. Missing a Picking metric — flagged on page 5.

Immediate candidates for the next session

Live options on the table — not a forced order.

1. **Dispatch** — the last piece of the Outbound/Picking/Dispatch trio. Needs its own align-before-coding pass: whether an order can dispatch with non-priority lines still short as long as priority lines are complete, what the Dispatch action itself is (a new explicit action vs. piggybacking on the existing Gate Out step), and whether load needs its own re-scan or can trust the picking ledger.
2. ~~Close the two real Picking gaps (no claim expiry, NEEDS_SOURCE retry)~~ — **NEEDS_SOURCE retry BUILT 2026-09-15**. Claim-expiry for an abandoned `PickTrip` still open (Putaway already solved this exact problem, `PutawayClaimExpiryScheduler`).
3. ~~Wire Picking into Pick Face~~ — **BUILT 2026-09-15**. `suggestPickPosition()` now tries Pick Face before reserve.
4. ~~Operator assignment fairness for Picking~~ — **BUILT 2026-09-15**. Mirrors Putaway's idle-time ranking + Supervisor/Manager escalation.
5. Add a Picking metric to Analytics — Marrying/Putaway/Pick Face are all reported per operator; `PickTrip` isn't. **Still open**.
6. The reslotting/consolidation suggestion engine — genuinely unblocked (ABC, dock-relative placement, and FMS are all in place). Schema has sat ready since Topic 1.
7. Daily Outward Volume (Analytics) — the mirror of the already-built Daily Inward Volume chart. Now genuinely unblocked (Picking writes real PICK movements) — still needs a design conversation on exactly which movement(s) represent the warehouse-exit event.
8. ~~Exception/short pick handling mid-pick~~ — **DESIGNED & BUILT 2026-09-15** (auto short-quantity detection + an explicit "Report a Problem" action, both routing to Supervisor+ review).
9. Pick Face for Drive-in / Ground / Stillage — SPR-only today, deliberately.
10. Dock-out → Gate-out signal, Dock + Staging + Yard visualizer, a cancel/void path for a mistaken Gate In — all previously flagged, still not designed.

Known, accepted v1 simplification (2026-09-15, flagged not silently assumed acceptable forever): retry/recovery only ever excludes the single location that most recently failed, not a running history of every location ever tried for a task — unlike Putaway's `PutawayReassignment` table for "Request Different Bin." Not building a `PickSourceReassignment` history table for this yet.

Deferred, lower priority — per explicit client calls, don't build unprompted

From the top-WMS gap checks (Yard/Gate 2026-09-07, Picking 2026-09-14 — see page 5)

- Wave / batch / zone / cluster picking — "basics first" confirmed; single-task discrete picking only, for now.
- Task Interleaving — the one exception on this list: flagged HIGH PRIORITY now that Picking genuinely exists, not just "someday."
- Cycle counting / physical inventory reconciliation — no workflow exists at all yet.
- 3PL / client billing — a real gap against the multi-company SaaS goal; no rate cards or invoicing today.
- Vendor/Supplier master data — receiving is tied to a Vehicle, not to a vendor.
- Kitting/Assembly logic — schema exists, zero logic built.
- Quality Control / Inspection workflow — no formal hold/inspect/release/reject module.
- Cartonization / pack-station / sortation — likely an e-commerce-parcel concept; may not apply to our pallet/case B2B model. Flagged, not assumed either way.
- Voice picking / pick-to-light — hardware investments, not software gaps.

Putaway follow-ons

- A cancel/exception path for a task that can't be completed.
- Correcting an already-completed mis-putaway.
- Real queue-ordering / aging-based task prioritization.
- Pick Face for Drive-in / Ground / Stillage (SPR-only today, deliberately).
- Stillage's "closed column" lifecycle tracking — unblocked now that Picking writes `real MovementType.PICK` movements, not yet picked up.
- Pallet reuse-on-depletion — now genuinely exercisable (Picking writes PICK), not yet wired/tested.

Picking follow-ons (2026-09-15)

- Claim-expiry for an abandoned `PickTrip` — Putaway's own `PutawayClaimExpiryScheduler` pattern, not yet ported to Picking.
- A running history of every source location tried per task (unlike Putaway's `PutawayReassignment`) — today's retry only excludes the single most-recent failure.

Everything else explicitly parked

- Self-service driver check-in, Yard Plan View, closing a short receipt, Batch/Lot + expiry capture.
- Real GS1/barcode parsing, camera-based scanning, real SMS/Email/WhatsApp/Voice providers.
- Dock Scheduling algorithm, blacklist enforcement at Gate In, WMS/TMS integration.
- SUPER_ADMIN creation, a configurable permission matrix, an invite-link flow.
- Inventory Control Policy, Opening Balance load, dispatch-proximity distance calc, real SAP/ERP integration, cloud deployment.

Picking — gap analysis vs. other WMS platforms (2026-09-14, statuses updated 2026-09-15)

Researched against Manhattan Active WM, SAP EWM, and Blue Yonder, then checked directly against our own code — each row is a confirmed gap, a capability needing its own design pass, or something already deliberately deferred by client call.

Gap	Type	What other WMS platforms do	Status in our system	Priority
No claim-expiry for an abandoned pick trip	Real gap	An RF/scan-driven claim that's never completed times out and returns to the pool automatically.	Still open. Putaway already solved this exact problem (<code>PutawayClaimExpiryScheduler</code> , 30-min auto-expiry). Picking has no scheduler at all — a claimed-then-abandoned <code>PickTrip</code> stays <code>IN_PROGRESS</code> forever, permanently blocking that task.	High
NEEDS_SOURCE task has no retry path	Real gap	A short-stock/no-source condition routes to an alternate-location search or a Supervisor exception queue.	BUILT 2026-09-15. New <code>attemptSourceRecovery()</code> + <code>retryTaskSource()</code> + a Retry Source button. A dead-end task's quantity is trimmed and a sibling <code>NEEDS_SOURCE</code> task spun off for any leftover — both auto-logged with a <code>SHORT_STOCK</code> exception.	Built
Picking doesn't use Pick Face	Real gap	Fast-access forward-pick locations are always preferred over reserve storage for an eligible SKU.	BUILT 2026-09-15. <code>suggestPickPosition()</code> is now a two-phase wrapper — Pick Face (SPR) is always tried first, falling back to reserve only when empty for that SKU.	Built
No operator-fairness layer for Picking	Real gap	Idle-time-based task assignment recommendations, with supervisor escalation if work backs up.	BUILT 2026-09-15. Mirrors Putaway's own layer: idle-time ranking, a live recommendation banner, a new <code>PickAssignmentScheduler</code> (Supervisor alert, then Manager escalation).	Built
No Picking metric in Analytics	Real gap	Per-operator trip time / productivity reported for every pick-type activity.	Still open. <code>AnalyticsService</code> reports Marrying, Putaway, and Pick Face operator productivity — nothing for <code>PickTrip</code> yet. An abandoned-claim list is also still pending (meaningful once claim-expiry exists).	Medium
No mid-pick exception handling	Needed design	An operator can flag "correct location, item genuinely not there" or "damaged," routing to a Supervisor — same shape as a blocked Inbound scan.	DESIGNED & BUILT 2026-09-15. Short quantity is now auto-detected on completion (compared against the trip's own claimed amount, not the task's total, so equipment-capacity-capped trips never false-trigger); a new "Report a Problem" action lets an operator explicitly flag Empty Bin / Wrong SKU / Damaged / Other, abandoning the trip and recovering a new source. Supervisor+ get a review queue (new <code>PICK_EXCEPTION_REVIEW_ROLES</code>).	Built
Wave / batch / zone / cluster picking	Deferred	Industry-standard picking orchestration — batching/waving orders together cuts travel 30-60% at scale.	Explicitly deferred — "basics first." Today's model is single-order, single-task discrete picking only.	Confirmed deferred
Task Interleaving (Putaway + Picking)	Deferred	Combines a putaway trip and a pick trip into one path to cut empty travel.	Already flagged HIGH PRIORITY in this roadmap for "the moment Picking exists" (2026-09-07) — that moment is now. Still not designed.	High (revisit now)
Cartonization / pack-station / sortation	Flag only	Optimal box selection and multi-item order consolidation before shipping — common in e-commerce-parcel WMS.	Not built, and may not apply — our model is pallet/case-level B2B fulfillment with a fixed per-order staging bin, not parcel consolidation. Worth a quick confirm, not assumed either way.	Confirm applicability
Voice picking / pick-to-light	Hardware	Hands-free picking technology — 25-50% productivity gains, >99% accuracy in mature deployments.	Not applicable to the software layer — a hardware investment decision, not a code gap. Flagged for awareness only.	Out of scope

Verified this pass via a throwaway-company diagnostic script invoking the real, unmodified `PickTasksService/PickAssignmentScheduler` methods directly (23/23 checks). Sources: Manhattan Active WM (manh.com), SAP EWM Picking Strategies (help.sap.com), Blue Yonder WMS (info.blueyonder.com), RFgen Warehouse Picking Guide, Lucas Systems Order Picking Productivity, Cleverence Warehouse Replenishment.